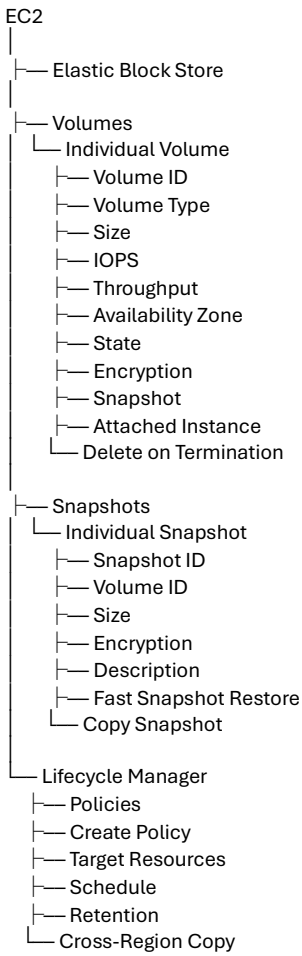
Amazon EBS — Console Dashboard Wise Important Interview Questions & Answers



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1. What is Amazon EBS?
Answer:
Amazon EBS stands for Elastic Block Store and provides block storage for EC2 instances.
It works like a virtual hard disk that can be attached to an EC2 instance.
We can store operating system files, applications, databases, and other persistent data on EBS.
For example:
Suppose we launch an EC2 server and install a database on it.
We can attach an EBS volume and store the database data on that volume.
Even if the EC2 instance is stopped, the EBS volume can keep the data.
Example:
EC2
↓
EBS Volume
↓
Application / Database Data
Simple ga gurthupettuko:
EBS = EC2 ki virtual hard disk

EC2 → Elastic Block Store → Volumes
2. What is an EBS Volume?
Answer:
An EBS Volume is a block storage device that can be attached to an EC2 instance.
It provides persistent storage for operating system files, applications, and data.
A volume is created in a specific Availability Zone.
For example:
Suppose our EC2 instance needs additional storage for application files.
We create a 100 GB EBS volume in the same Availability Zone and attach it to EC2.
The application can then use that volume like a disk.
Example:
EC2 Instance
↓

100 GB EBS Volume
↓
Application Files
Simple ga gurthupettuko:
Volume = Virtual disk

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3. What is the Volume Type in EBS?
Answer:
Volume Type defines how an EBS volume performs and what workload it is suitable for.
Common types include General Purpose SSD, Provisioned IOPS SSD, Throughput Optimized HDD, and Cold HDD.
We choose the type based on performance, workload, and cost requirements.
For example:
For a normal application server, we may use gp3.
For a database requiring high and predictable IOPS, we may choose an io2 volume.
Example:
Application
↓
EBS Volume
↓
Volume Type
├— gp3
├— io2
├— st1
└— sc1
Simple ga gurthupettuko:
Volume Type = Performance + Cost choice

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4. What is gp3 in EBS?
Answer:
gp3 is a General Purpose SSD EBS volume type designed for a wide range of workloads.
It allows us to configure storage size, IOPS, and throughput independently within its supported limits.
It is commonly used for EC2 operating systems and general-purpose applications.
For example:
Suppose our application needs 100 GB storage but also needs more IOPS than the default performance.
We can use gp3 and configure the required IOPS without necessarily increasing the storage size.
Example:
EC2
↓
gp3
├— Size
├— IOPS
└— Throughput
Simple ga gurthupettuko:
gp3 = General-purpose SSD

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5. What is io2 in EBS?
Answer:
io2 is a Provisioned IOPS SSD volume designed for workloads that need high and consistent I/O performance.
It is commonly considered for demanding databases and critical applications.
We can provision IOPS according to the workload requirement within the supported limits.
For example:
Suppose a production database performs many read and write operations and requires predictable storage performance.
We can use an io2 volume and provision the required IOPS.
This provides more predictable performance than a basic general-purpose configuration.
Example:
Production Database
↓
io2
↓
High / Predictable IOPS
Simple ga gurthupettuko:
io2 = High-performance SSD for demanding workloads

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6. What is the Size field in an EBS Volume?
Answer:
The Size field defines how much storage capacity the EBS volume provides.
We select the size based on how much data the application needs.
EBS volumes can generally be increased in size, subject to the volume type and service limits.
For example:
Suppose our application currently needs 100 GB but later grows to 200 GB.
We can modify the EBS volume and increase its size instead of creating a completely new disk.

Example:
EBS
↓
100 GB
↓
Modify
↓
200 GB
Simple ga gurthupettuko:
Size = Storage capacity

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7. What is IOPS in EBS?
Answer:
IOPS means Input/Output Operations Per Second.
It represents how many read or write operations the storage can handle per second.
Higher IOPS can be important for workloads that perform many small read and write operations.
For example:
Suppose a database performs thousands of small read and write operations every second.
It may need higher IOPS than a simple file server.
We select the EBS configuration according to that workload.
Example:
Database
↓
EBS
↓
High IOPS
↓
Fast Read / Write Operations
Simple ga gurthupettuko:
IOPS = Number of storage operations per second

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8. What is Throughput in EBS?
Answer:
Throughput represents how much data can be transferred through the volume per second.
It is commonly measured in MiB/s.
Throughput is especially important for workloads that process large amounts of sequential data.
For example:
Suppose an application processes large files continuously.
It may benefit from higher throughput because large amounts of data need to be transferred efficiently.
Example:
Large Files
↓
EBS
↓
High Throughput
↓
Fast Data Transfer
Simple ga gurthupettuko:
Throughput = Amount of data transferred per second

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9. What is the Availability Zone of an EBS Volume?
Answer:
An EBS volume is created in a specific Availability Zone.
Normally, the volume can be attached directly to EC2 instances in the same Availability Zone.
This is an important consideration when creating or attaching EBS volumes.
For example:
Suppose an EBS volume is created in AZ-1a.
An EC2 instance in AZ-1b normally cannot directly attach that volume.
We need the volume and instance to be in compatible locations.
Example:
Region
↓
AZ-1a
├── EC2
└── EBS
Simple ga gurthupettuko:
EBS Volume = AZ-specific

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10. Can we attach an EBS Volume to an EC2 instance in another Availability Zone?
Answer:
Normally, an EBS volume is attached to an EC2 instance in the same Availability Zone.

EBS volumes are AZ-specific, so they cannot simply be attached across AZs.
If we need the data in another AZ, we can use a snapshot and create a new volume there.
For example:
Suppose our volume is in AZ-1a but we want storage in AZ-1b.
We can create a snapshot of the volume and create a new EBS volume from that snapshot in AZ-1b.
Example:
AZ-1a
EBS Volume
↓
Snapshot
↓
AZ-1b
New EBS Volume
Simple ga gurthupettuko:
Different AZ = Snapshot → New Volume

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11. What is Encryption in EBS?
Answer:
EBS encryption protects data stored on the volume and associated snapshots.
It uses AWS Key Management Service keys or supported AWS-managed keys to encrypt the data.
Encryption helps protect sensitive application and database data.
For example:
Suppose an EC2 server stores customer information on EBS.
We can create an encrypted EBS volume so the stored data is protected.
The related snapshots can also be encrypted.
Example:
EC2
↓
Encrypted EBS
↓
Customer Data
🔐
Simple ga gurthupettuko:
EBS Encryption = Protect data at rest

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12. What is the Attached Instance field?
Answer:
The Attached Instance field shows which EC2 instance is currently using the EBS volume.
It helps us understand where the volume is connected.
This information is useful when managing or troubleshooting storage.
For example:
Suppose we have an EBS volume called data-volume.
The console can show that it is attached to EC2-Server-01.
We can then identify which server is using that storage.
Example:
EBS Volume
↓
Attached To
↓
EC2-Server-01
Simple ga gurthupettuko:
Attached Instance = Which EC2 is using the volume

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13. What is Delete on Termination?
Answer:
Delete on Termination determines whether an EBS volume is automatically deleted when its attached EC2 instance is terminated.
The setting can be configured differently for different volumes.
We should be careful with this setting when the volume contains important data.
For example:
Suppose the root volume contains only the operating system.
We may allow it to be deleted when EC2 is terminated.
For an important data volume, we may keep it after EC2 termination.
Example:
EC2 Terminated
↓
┌──────────┐
↓ ↓
Delete Keep
Root Disk Data Disk
Simple ga gurthupettuko:
Delete on Termination = EC2 terminate ayyaka disk em avvali?

EC2 → Elastic Block Store → Snapshots

14. What is an EBS Snapshot?

Answer:

An EBS Snapshot is a point-in-time backup of an EBS volume.
It can be used to restore data or create new EBS volumes.
Snapshots are stored in Amazon S3-managed infrastructure rather than being directly accessed as normal S3 objects.

For example:

Suppose our EC2 server has an important data volume.
Before making a major change, we can create a snapshot.
If something goes wrong, we can use that snapshot to create a new volume.

Example:

EBS Volume
↓
Snapshot
↓
New EBS Volume

Simple ga gurthupettuko:

Snapshot = Point-in-time backup of EBS

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15. Why do we use EBS Snapshots?

Answer:

We use snapshots mainly for backup, recovery, migration, and creating new EBS volumes.
They provide a way to preserve the state of a volume at a particular point in time.
Snapshots can also be copied to another Region for disaster recovery.

For example:

Suppose we want to upgrade an application and are worried about data loss.
We create an EBS snapshot before the upgrade.
If required, we can restore the data from that snapshot.

Example:

EBS
↓
Snapshot
↓
Backup / Recovery

Simple ga gurthupettuko:

Snapshot = Backup + Recovery

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16. What is Incremental Snapshot?

Answer:

After the first snapshot, subsequent EBS snapshots are incremental.
Only the changed blocks since the previous snapshot need to be stored as additional snapshot data.
This helps make repeated backups more storage-efficient.

For example:

Suppose we create the first snapshot of a 100 GB volume.
Later, only 10 GB of data changes.
The next snapshot tracks the changed blocks rather than treating the entire volume as completely new data.

Example:

Snapshot 1
↓
Changed Blocks
↓
Snapshot 2
↓
Changed Blocks
↓
Snapshot 3

Simple ga gurthupettuko:

Incremental = Only changed data after previous snapshot

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17. Can we create a new EBS Volume from a Snapshot?

Answer:

Yes, we can create a new EBS volume from an EBS snapshot.
The new volume can be created in the Availability Zone where we need to use it.
This is useful for recovery, migration, and creating additional copies of storage.

For example:

Suppose an original EBS volume is unavailable but we have its snapshot.
We can create a new volume from that snapshot and attach it to another EC2 instance.

Example:

Snapshot
↓

Create Volume
↓
New EBS Volume
↓
EC2
Simple ga gurthupettuko:
Snapshot → New Volume

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18. What is Copy Snapshot?
Answer:
Copy Snapshot allows us to create a copy of an EBS snapshot.
The copy can be created in another AWS Region, which is useful for disaster recovery.
We can also use snapshot copies for migration and independent backup management.
For example:
Suppose our production EBS snapshot is in Mumbai.
We can copy the snapshot to another Region.
If the primary Region has a major issue, the copied snapshot can help us create storage there.
Example:
Mumbai
EBS Snapshot
↓
Copy Snapshot
↓
Another Region
↓
New Volume
Simple ga gurthupettuko:
Copy Snapshot = Snapshot ni another Region/account setup ki copy cheyyadam

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19. What is Fast Snapshot Restore?
Answer:
Fast Snapshot Restore, or FSR, helps make volumes created from a snapshot fully initialized faster in selected Availability Zones.
Without it, blocks may need to be initialized as they are accessed, which can affect initial performance.
FSR is useful when we need fast access to restored volumes.
For example:
Suppose we frequently restore an EBS volume from a snapshot during disaster recovery.
We can enable Fast Snapshot Restore for the required Availability Zone.
The restored volume can then avoid the normal lazy initialization behavior.
Example:
EBS Snapshot
↓
Fast Snapshot Restore
↓
New Volume
↓
Ready for workload
Simple ga gurthupettuko:
FSR = Snapshot nundi volume fast ga ready cheyyadam

EC2 → Elastic Block Store → Lifecycle Manager
20. What is EBS Lifecycle Manager?
Answer:
EBS Lifecycle Manager helps automate the creation and management of EBS snapshots and related backups.
We create policies that define which resources to target, when snapshots should be created, and how long they should be retained.
This reduces the need to create snapshots manually.
For example:
Suppose production EC2 servers need daily EBS backups.
We can create a Lifecycle Manager policy to automatically create snapshots every day.
We can also configure retention so old snapshots are removed automatically.
Example:
EBS Volume
↓
Lifecycle Manager
↓
Daily Snapshot
↓
Retention
↓
Old Snapshot Removed
Simple ga gurthupettuko:
Lifecycle Manager = Automatic EBS backup management

21. What are EBS Lifecycle Manager Policies?
Answer:
A Lifecycle Manager policy defines how EBS snapshots or related resources should be managed automatically.
The policy can specify target resources, schedule, retention, and other supported actions.
It helps create a consistent backup process.
For example:
Suppose we want snapshots every 24 hours and want to retain only the latest seven snapshots.
We can create a Lifecycle Manager policy with that schedule and retention configuration.
Example:
Lifecycle Policy
├— Target
├— Schedule
└— Retention
↓
Automatic Snapshots
Simple ga gurthupettuko:
Policy = Backup rules

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22. What is Retention in EBS Lifecycle Manager?
Answer:
Retention defines how long or how many generated snapshots should be kept according to the lifecycle policy.
After the configured retention condition is reached, older snapshots can be removed automatically.
This helps control storage costs and keeps backups manageable.
For example:
Suppose we create one snapshot every day and configure retention for seven snapshots.
The system keeps the required recent snapshots and removes older ones according to the policy.
Example:
Day 1 → Snapshot
Day 2 → Snapshot
...
Day 8 → New Snapshot
↓
Old Snapshot Removed
Simple ga gurthupettuko:
Retention = How long/how many backups keep cheyyali

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23. What is the difference between EBS Snapshot and AMI?
Answer:
An EBS Snapshot is a point-in-time backup of an EBS volume.
An AMI is an image used to launch EC2 instances and can include the required root volume information and other configuration.
A snapshot is mainly storage-level backup, while an AMI is designed for launching EC2 instances.
For example:
Suppose we want to back up only a data disk.
We can create an EBS snapshot.
If we want to create new EC2 instances with a prepared operating system and application setup, we can create an AMI.
Example:
EBS Volume
↓
Snapshot
↓
Backup

EC2
↓
AMI
↓
New EC2 Instance
Simple ga gurthupettuko:
Snapshot = Disk backup | AMI = EC2 launch image

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24. What happens to EBS data when an EC2 instance is stopped?
Answer:
When an EC2 instance is stopped, its attached EBS volumes normally remain available and retain their data.
The EBS volume is persistent storage, unlike instance store storage.
When the instance is started again, the EBS volume can be used again.
For example:
Suppose our EC2 server has an EBS volume containing application data.
We stop the EC2 instance at night to save compute cost.
When we start it again, the EBS volume and its data remain available.
Example:

EC2 Running
↓
EBS Data
↓
EC2 Stopped
↓
EBS Data Remains
↓
EC2 Started
Simple ga gurthupettuko:
EC2 Stop ≠ EBS data delete

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25. What happens to EBS when an EC2 instance is terminated?
Answer:
It depends on the Delete on Termination setting of the EBS volume.
If Delete on Termination is enabled, the volume is deleted when the EC2 instance is terminated.
If it is disabled, the volume remains and can be managed separately.
For example:
Suppose an EC2 instance has a root volume and a separate data volume.
We may allow the root volume to be deleted with the instance but keep the data volume for future use.
Example:
EC2 Terminate
↓
┌──────────┐
↓ ↓
Root Volume Data Volume
Delete Keep
Simple ga gurthupettuko:
Terminate → Check Delete on Termination

Important EBS Console Questions
26. What important things do you check in the EBS Volumes console?
Answer:
In the Volumes console, we commonly check volume type, size, IOPS, throughput, Availability Zone, state, encryption, and attached instance.
We also check whether the volume is attached or available and whether it is encrypted.
These details help us understand the storage configuration and troubleshoot issues.
For example:
If an application is slow, we can check the volume type, IOPS, throughput, and attachment information.
If the volume is not attached, we can check its Availability Zone and state.
Example:
EBS Volume
├— Type
├— Size
├— IOPS
├— Throughput
├— AZ
├— Encryption
└— Attached EC2
Simple ga gurthupettuko:
Volume Console = Performance + Location + Security + Attachment

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27. What important things do you check in the EBS Snapshots console?
Answer:
In the Snapshots console, we commonly check snapshot ID, source volume, size, encryption, creation time, and description.
We can also manage snapshot copies and supported restore-related options.
These details help us verify whether the required backup exists and can be used for recovery.
For example:
Before an application migration, we can check whether a recent snapshot exists.
If the snapshot is available, we can use it to create a new volume for the migration.
Example:
Snapshots
├— Snapshot ID
├— Source Volume
├— Size
├— Encryption
└— Created Time
Simple ga gurthupettuko:
Snapshot Console = Check and manage EBS backups
